

1. We first need to show that LONGEST SIMPLE CYCLE is in NP. An instance x is an undirected graph and an integer k . Let a certificate y be a sequence of at least k vertices from the graph. The verifier is similar to that for traveling salesman: it goes through the vertices in the certificate one by one, marking each as it goes to check that there are no repeats, and verifying that there is an edge from each vertex in the certificate to the next one given in the certificate. Finally, the verifier checks that the cycle is completed by an edge from the last vertex in the certificate back to the first vertex. If these checks are satisfied, the verifier outputs 1. If any fail, the verifier outputs 0.

This verifier takes linear time in the number of vertices. If there is a simple cycle with at least k vertices, then passing that cycle as a certificate causes the verifier to output 1. Otherwise, no certificate contains a valid cycle with at least k vertices, so the verifier always outputs 0 for this instance. Since the verifier is correct and takes polynomial time, the problem is in NP.

To show the problem is NP-hard, we reduce HAM-CYCLE to it. Given a graph G that is an instance of HAM-CYCLE, define an instance $\langle G, k \rangle$ of LONGEST SIMPLE CYCLE containing the same graph $G = (V, E)$, and $k = |V|$, the number of vertices in G .

This reduction involves only simple copying and counting, so it takes polynomial time.

Now, if graph G contains a Hamiltonian cycle, then this cycle is a simple cycle with $|V|$ vertices, so the correct decision on the instance of LONGEST SIMPLE CYCLE is 1. If there is no Hamiltonian cycle, then there is no simple cycle with $|V|$ or more vertices (simply by the definition of Hamiltonian cycle), so the correct decision on the instance of LONGEST SIMPLE CYCLE is 0.

So, the reduction is correct and takes polynomial time, showing that LONGEST SIMPLE CYCLE is NP-hard. Since it is also in NP, it is NP-complete.